

Dissection of Maize Kernel Composition and Starch Production by Candidate Gene Association

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Cereal starch production forms the basis of subsistence for much of the world's human and domesticated animal populations. Starch concentration and composition in the maize (*Zea mays ssp mays*) kernel are complex traits controlled by many genes. In this study, an association approach was used to evaluate six maize candidate genes involved in kernel starch biosynthesis: *amylose extender1* (*ae1*), *brittle endosperm2* (*bt2*), *shrunk1* (*sh1*), *sh2*, *sugary1*, and *waxy1*. Major kernel composition traits, such as protein, oil, and starch concentration, were assessed as well as important starch composition quality traits, including pasting properties and amylose levels. Overall, *bt2*, *sh1*, and *sh2* showed significant associations for kernel composition traits, whereas *ae1* and *sh2* showed significant associations for starch pasting properties. *ae1* and *sh1* both associated with amylose levels. Additionally, haplotype analysis of *sh2* suggested this gene is involved in starch viscosity properties and amylose content. Despite starch concentration being only moderately heritable for this particular panel of diverse maize inbreds, high resolution was achieved when evaluating these starch candidate genes, and diverse alleles for breeding and further molecular analysis were identified.

INTRODUCTION

As a result of increased demands on food production from escalating population growth and environmental degradation, interest in improved breeding strategies for agricultural crops is growing. Progress in cereal starch production is especially important because these starches comprise 55 to 75% of daily human food intake and are the main source of food for domestic animals (Pan, 2000). In addition to being the largest production crop in the world (<http://apps.fao.org>), maize (*Zea mays ssp mays*) has numerous starch mutants that provide a unique source of specialty starches, including amylose-free *waxy*, important for many industrial applications (Lambert, 2001), and the *sugary* mutants, responsible for the production of popular sweet maize varieties (Pan, 2000). Interest is also growing in the use of maize as fuel, marked by recent breeding attempts to enhance efficiency in fermentation to ethanol (Dien et al., 2002). Thus, identifying those genes and alleles in maize that control traits such as grain yield, starch concentration, and starch quality is an

important step in meeting the future goals of both agriculture and industry.

Research on well-known mutants of maize has helped elucidate key genes involved in the starch pathway. Sucrose transported into the maize kernel is converted to UDP-glucose and fructose by the major isoform of sucrose synthase, encoded by the *shrunk1* (*sh1*) gene (Chourey and Nelson, 1976). *sh2* and *brittle endosperm2* (*bt2*) encode the large and small subunits, respectively, of ADP-glucose pyrophosphorylase (AGPase), which converts ADP-glucose into glucose-1-phosphate, the substrate for starch synthases (Tsai and Nelson, 1966; Bae et al., 1990; Bhavé et al., 1990). Generally regarded as the rate-limiting step in starch biosynthesis, AGPase is allosterically regulated by 3-phosphoglycerate and Pi and thus is a target for controlling starch yield through the modification of its allosteric effector sites (Stark et al., 1992).

Starch synthases then sequentially add glucose-1-phosphate molecules onto the nonreducing ends of a growing starch chain. Granule-bound starch synthase, encoded by the *waxy1* (*wx1*) locus in maize, is solely responsible for amylose production (Nelson and Rines, 1962; Shure et al., 1983). Amylose-free starches caused by the *wx1* mutant in maize have long been of commercial interest (Deatherage et al., 1954). Mutants of a second gene, *amylose extender1* (*ae1*), result in maize kernels with higher amounts of amylose than nonmutant kernels (Fisher et al., 1996; Kim et al., 1998). The *ae1* gene codes for the starch branching enzyme IIb isoform, which hydrolyzes $\alpha(1 \rightarrow 4)$ linkages and reattaches these chains with $\alpha(1 \rightarrow 6)$ branch points

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found in amylopectin. *sugary1* (*su1*) encodes a debranching enzyme of the isoamylase type. Mutant *su1* kernels contain the highly branched, water-soluble phytylglycogen and constituted the original sweet corns (James et al., 1995). To obtain the semicrystalline formation of amylopectin, it may be that the correct ratio of starch branching to debranching enzymes is important (Ball et al., 1996), but the role of isoamylase in conjunction with branching enzymes has not yet been resolved, and differing models have been proposed (reviewed in Smith, 2001).

Although scientists understand the basic structure of the molecule itself, there is still much to be learned about starch metabolism and the organization of the starch granule. Maize starch is composed of 21% amylose, a mostly linear chain of $\alpha(1 \rightarrow 4)$ linked glucose molecules, and 79% amylopectin, a more highly branched molecule of $\alpha(1 \rightarrow 4)$ linkages with $\alpha(1 \rightarrow 6)$ branch points. Such high amylopectin content accommodates the long-term storage of starch in the kernel endosperm, where its semicrystalline nature allows for efficient packaging into granules. Attempts at producing amylopectin *in vitro*, however, have been unsuccessful, producing instead an animal-like glycogen product (Guan et al., 1995). In addition to the exact nature of amylopectin formation, several enzymes involved in starch biosynthesis are also poorly understood because many of these enzymes have multiple isoforms (Fisher et al., 1996; Gao et al., 1996; Huang and Wang, 1998; Sidebottom et al., 1998; Beckles et al., 2001).

Although geneticists and biochemists have identified many relevant genes, the ability to modify starch products for future gain will depend on an increased level of understanding regarding specific alleles that modify starch concentration and composition. Linkage mapping has identified several regions in the maize genome that have an effect on kernel starch concentration, with some of these quantitative trait loci (QTL) corresponding to starch biosynthesis genes. One of the most documented genes, *sh2*, colocalizes with a QTL effect on protein and starch levels (Goldman et al., 1993) and correlates with amylose levels (Prioul et al., 1999; Séne et al., 2000). Another positional candidate gene, *sh1*, has been linked to starch and protein concentration (Berke and Rocheford, 1995). Based on mutational studies, *bt2*, *ae1*, *su1*, *sh1*, *sh2*, and *wx1* are functional candidate genes thought to affect either starch concentration or starch composition quality. Mutant kernels for all six genes (individually) contain severely reduced starch concentrations or altered amylose/amylopectin ratios. As such, all six genes provide good starting points in the quest for functional polymorphisms that directly affect starch metabolism in maize.

Although the aforementioned QTL studies are initially suggestive, their resolution is often on the order of 10 centimorgans (cM) or more. Because this distance can correspond to millions of bases in maize, real evaluation of individual genes has not occurred. By contrast, association mapping can provide a high-resolution alternative for the evaluation of these candidate genes and has the potential to evaluate a wide range of alleles (Buckler and Thornsberry, 2002; Flint-Garcia et al., 2003). Though common in human genetics (Lander and Schork, 1994; Risch and Merikangas, 1996), association approaches have only recently been applied to plant populations (Flint-Garcia et al., 2003). The comparatively high resolution provided by association mapping

is dependent upon the amount of linkage disequilibrium (LD), or the nonrandom association of alleles, present in a species. Two LD studies in maize for both diverse inbreds as well as traditional landraces suggest that in most cases LD decays rapidly within genes, usually within 2000 bp (Remington et al., 2001; Tenaillon et al., 2001). Therefore, high-resolution association mapping is possible in maize. This resolution can be reduced when genes have been recent targets of selection, as has been the case for several genes in the starch pathway (Whitt et al., 2002) and kernel carotenoids (Palaisa et al., 2003).

One difficulty, however, in applying association methods is that LD can be present as a result of genetic drift, selection, or population admixture. Thus, as sometimes seen in human populations, LD can contain the confounding effect of population substructure, resulting in a high frequency of false positive associations (Lander and Schork, 1994). To control for this type of structure, Pritchard et al. (2000a) developed a statistical approach that assigns membership to various subpopulations by determining the amount of genotypic correlation based on unlinked, random markers. Thornsberry et al. (2001) adapted Pritchard's approach for use with quantitative variation and then successfully applied it to the evaluation of maize flowering time. By including estimates of population structure in this analysis, the risk of obtaining false positive associations was reduced (Thornsberry et al., 2001).

Interestingly, these population structure estimation procedures—initially designed for outbred populations—are useful for evaluating these diverse maize lines for several reasons (Remington et al., 2001; Thornsberry et al., 2001; Liu et al., 2003). First, most of these inbreds are unrelated to one another because they are essentially derived from extremely outbred landraces or from synthetic populations. Closely related pairs of lines were also eliminated when first choosing a diverse maize panel for this study. Second, many of the breeding crosses were essentially random. Finally, the actual inbreeding process is not modeled or relevant to these population structure estimates because the genotypes are treated as haplotypes for analysis (Falush et al., 2003).

In this study, six maize candidate genes involved in kernel starch biosynthesis (*ae1*, *bt2*, *sh1*, *sh2*, *su1*, and *wx1*) were tested for associations with starch concentration and starch composition quality using the structured association method of Thornsberry et al. (2001). Each gene was sequenced in a diverse set of maize inbreds, a germplasm that captured 80 to 90% of the microsatellite diversity found in maize landraces (Liu et al., 2003). The use of these maize inbreds reduced the analysis to essentially one haplotype per line, allowing for examination of additive effects only. By locating those allelic regions associated with either starch concentration or composition, polymorphisms identified in this survey can be used in future genetic and breeding studies to manipulate these important agronomic traits.

RESULTS

Principal Component Analysis

Phenotypic trait values for kernel composition (near infrared [NIR]) and starch pasting traits from summer (Clayton, NC 2001;

summer^C) and winter (Homestead, FL 1998; winter^H) replications were analyzed separately using principal component analysis (PCA) on the covariance matrix of traits. PCA takes complex correlated data arranged in multidimensional space and reduces the high dimensionality of the data into more simple, linearized axes while retaining as much of the original variation as possible. All correlated components of sample data will form a correlation matrix, where the variances of the transformed, standardized data along an axis (eigenvectors) are the principal components. Such axes correspond to the largest eigenvalues in the direction of the largest variation of the data. PCA was used in this study to reduce multiple testing in the association analyses by summarizing the phenotypes over the various replications and by combining correlated traits into single PCA indexes. PCA is appropriate for these kernel data, where protein, oil, and starch compositions in cereals are correlated traits (Dudley and Lambert, 1992, 2004).

PCA results for kernel composition along with subjective interpretations of eigenvectors for each factor are included in Table 1. Cumulatively, three factors explained 55% of the variation in phenotypes, where factor one alone explained 34%. Table 1 also shows the results of PCA for starch pasting values of the winter^H and summer^C field seasons. Cumulatively, three factors explained 91 and 94% of the variance in pasting and viscosity traits seen in maize kernels from the winter^H and summer^C environments, respectively.

Extent of LD in Starch Genes

Levels of LD decay varied across the six starch genes, as also seen in the Remington et al. (2001) study. Whereas LD declines rapidly for many genes in this diverse inbred panel (<2000 bp), three of the six starch genes studied here (*ae1*, *sh2*, and *su1*) had substantial LD present for 6000 to 20,000 bp (Figure 1). Thus, in these three cases, the resolution of association mapping is more likely to approach the gene level, rather than subgene resolution. All three genes with extensive LD have been recent targets of selection (Whitt et al., 2002).

Associations with Kernel Composition and Starch Pasting Properties

Of the six starch genes sampled, four showed significant associations ($P \leq 0.05$) for one or more traits (Table 2). Overall, *ae1* associated with pasting temperature (summer^C starch pasting factor three) and amylose content. *bt2* associated with oil versus protein production (kernel composition factor two). *sh1* showed an association with a general genotype $\times$ environment ($G \times E$) effect (kernel composition factor three) and with amylose content in the summer^C replication. Lastly, *sh2* associated with a general $G \times E$ effect (kernel composition factor three) and with starch viscosity characteristics (summer^C starch pasting factor one). Additionally, haplotype analysis of *sh2* indicated an effect on amylose content in the winter^H environment.

Associations that were less significant ($P \leq 0.10$) were also identified for those genes with high diversity (e.g., *sh1*, *sh2*, and *wx1*) and correspondingly low statistical power, but these will not be examined in detail (Table 2). The false discovery rate was estimated to be $\sim 50\%$ for associations where $P \leq 0.10$. When considering only the better replicated NIR data, then the estimated false discovery rate decreases to 21%. Overall, for associations where $P < 0.05$, only one or two false positives may exist.

We also empirically tested for false positives by associating the kernel composition NIR data with 10 genes in unrelated pathways (*d8*, *d3*, *zflA*, *zhd1*, *zmLD*, *fae2*, *ra1*, *tb1*, *zb7*, and *zb12*) using the three principal components for 30 tests. Only one test was significant at the 0.05 level (*d8* with PC2), which is slightly less than what is expected from chance. Even in this case, it is likely that the effects of *d8* on flowering translate into environmentally induced differences in grain filling. Such differences are a consequence of evaluating such a diverse panel of maize inbreds in any one environment, as these lines vary considerably in terms of photoperiod adaptations because of their native habitats. Thus, if these adaptive differences are taken into account, this empirical test resulted in no false positives, suggesting that most, if not all, of the starch associations identified in this study are a result of true linkage.

Table 1. Results of Kernel Composition and Each Starch Pasting PCA

Principal Component Factor	Proportion of Variance	Cumulative Variance	Interpretation ^a
NIR-1	0.34	0.34	Oil and protein accumulation versus starch
NIR-2	0.13	0.47	Oil production versus protein production
NIR-3	0.08	0.55	Genotype $\times$ environment effects
Starch Pasting-1 ^b	0.63/0.61	0.63/0.61	All viscosity traits and consistency ^c
Starch Pasting-2	0.20/0.23	0.83/0.84	Peak temperature ^d and peak time ^e
Starch Pasting-3	0.08/0.10	0.91/0.94	Pasting temperature

^a Interpretation of the trait(s) driving each principal component were determined by summing the absolute values of PCA eigenvectors (standardized values of the PCA covariance matrix) over all environments for each phenotypic trait. The greater the absolute value for a trait, the more weight that trait was given in the amount of its contribution to driving the variance seen for a particular PCA factor.

^b All starch pasting variance values are reported as winter^H/summer^C.

^c Gel consistency: cool paste viscosity minus hot paste viscosity (Pa-s).

^d Peak time: time required to reach peak (s).

^e Peak temperature: temperature of peak viscosity ($^{\circ}\text{C}$).

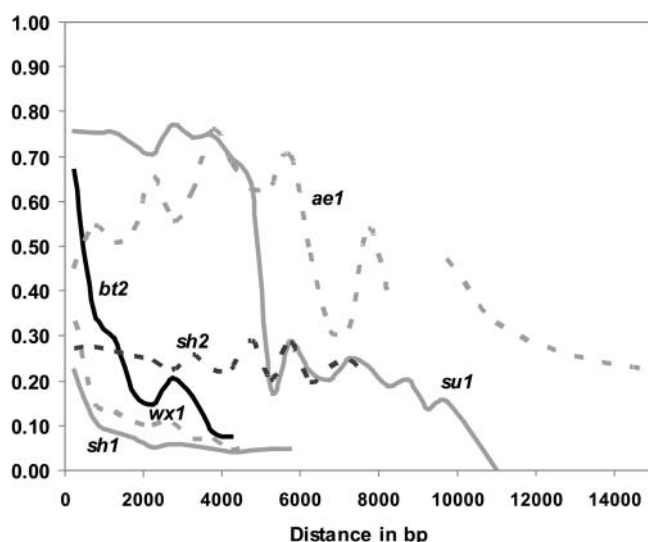


Figure 1. LD Plots of Squared Correlations of Allele Frequencies (r^2) against Distance between Polymorphic Sites for the Six Candidate Genes.

The lines indicate average r^2 for 500-bp windows for polymorphisms with a minimum frequency of 0.10. The *ae1* graph is based on 1000-bp windows because there were so few polymorphisms per window.

In the following detailed gene analyses, specific site(s) with the most significant results from overall gene testing are discussed; however, because of LD, associations with traits should be viewed as associations by haplotype, and not necessarily a specific polymorphism.

ae1

Overall, *ae1* significantly associated with pasting temperature (summer^C starch pasting factor three; Table 2) (logistic regression; $P = 0.02$). The most significant site was 1509, an adenosine to guanine transition located in exon two, hereafter referred to as *Ae1*-1509(G $\leftrightarrow$ A). The *Ae1*-1509(G) allele and associated haplotype were found in 13 lines out of the total 102, including both nonstiff stalk and semitropical lines (Table 3). Single nucleotide polymorphism (SNP) *Ae1*-1509(G) caused a nonsynonymous

change in the predicted AE1 protein sequence, converting Arg to Gly at amino acid 58 (R58G). The *ae1* orthologs in rice (*Oryza sativa*), wheat (*Triticum aestivum*), barley (*Hordeum vulgare*), and potato (*Solanum tuberosum*) all contain a Gly in the predicted amino acid sequences of the starch branching enzyme, whereas the majority of maize lines sampled in this study contained an Arg residue. The overall effect of *ae1* on pasting temperature (starch pasting factor three) was also seen in the winter^H replication, although at a marginal level of significance (Table 2). Taken alone, SNP *Ae1*-1509(G) showed a significant effect on starch pasting temperature in both summer and winter field replications (general linear model [GLM]; $P = 0.0002$ and 0.03 , respectively) (Figure 2). In both replications, the *Ae1*-1509(G) allele associated with 2.0 and 1.6% higher average pasting temperatures for summer^C and winter^H, respectively.

When summer^C and winter^H replications were analyzed simultaneously, *ae1* also showed an overall association with amylose content (logistic regression; $P = 0.03$). The most significant site was 1689, a SNP located in intron two containing a T $\leftrightarrow$ C transition. The polymorphic allele *Ae1*-1689(C) was in very strong LD with allele *Ae1*-1509(G) (LD; $r^2 = 0.9$) discussed above. In both summer and winter replications, lines containing the *Ae1*-1689C allele had 5.1 and 6.5% higher amylose content relative to those lines with the more common allele, *Ae1*-1689(T) (GLM; $P = 0.02$ and 0.004 , respectively) (Figure 1). Overall, this polymorphism explained 7 and 14% of the total variation in the two seasons, respectively.

bt2

Overall, *bt2* associated significantly with kernel composition factor two, which affected oil and protein levels (Table 2; logistic regression; $P = 0.05$). The most significant site was 925 located in exon one, in which a C $\leftrightarrow$ T transition has occurred (Figure 2). Within the ~ 400 bp sampled in the 102 lines, eight other SNPs and one insertion/deletion (indel) occurred in significant LD with site 925, revealing a haplotype. When considering the entire *bt2* gene alignment of 32 taxa, this distinct haplotype encompassed ~ 1000 bp at the 5' end of the gene. The SNP at site 925, hereafter referred to as *Bt2*-925(T), caused a nonsynonymous change in the N-terminal region of the BT2 protein, converting Pro to Leu at amino acid 22 (P22L). Whereas the mean oil content between lines varying for the *Bt2*-925(T) allele was not

Table 2. Results for Overall Gene Association Analyses Using Logistic Regression

Gene	Kernel Composition Factor			Starch Pasting Factor, Winter ^H			Starch Pasting Factor, Summer ^C			Amylose Content	
	1	2	3	1	2	3	1	2	3	Winter ^H	Summer ^C
<i>ae1</i>						*			**	*	
<i>bt2</i>		**									
<i>sh1</i>	*		**			*					**
<i>sh2</i>			**				**			*	
<i>su1</i>											
<i>wx1</i>											

*, ** = $P \leq 0.10$ and 0.05 , respectively. These are gene-wise P -values that account for multiple testing among sites. Exact P -values can be found in the text. Blank cells indicate $P > 0.10$.

Table 3. Maize Inbreds Surveyed and a Listing of Significant Alleles in the Association Tests

Maize Inbred	Subpopulation	Allele	Maize Inbred	Subpopulation	Allele
38-11	NSS		K55	NSS	C
A6 ^a	ST	B	Ki3	ST	B
A272 ^a	ST	B	Ki9 ^a	ST	B
A441-5	NSS	B, F, H	Ki11	ST	
A554	NSS		Ki21 ^a	ST	D, E
A619	NSS	C, D, E	Ki43	ST	B
A632	SS	C	Ki44	NSS	F, H
B14A ^a	SS	C	Ky21 ^a	NSS	C
B37 ^a	NSS	C	M37W	ST	
B68	SS		M162W	NSS	B
B73 ^a	SS		Mo17 ^a	NSS	B, D
B84	SS	C, D	Mo24W	NSS	D
B97 ^a	NSS	C	MS153	NSS	
B103 ^a	Mixed	C	N192	SS	C
B104	SS	C	N28Ht ^a	SS	C
C103	NSS	B, F, H	NC250	NSS	C
CI187-2 ^a	NSS	C, D	NC258	NSS	B
CM7	NSS	B	NC260 ^a	NSS	C
CM105	SS	C	NC296	ST	B
CM174	SS	C	NC298	ST	B
CML5	ST		NC300	ST	B
CML10	ST	F, H	NC304	ST	B
CML61	ST	F, H	NC320	ST	B, C
CML91	ST	C	NC338	ST	B
CML247	ST	C, E	NC348 ^a	ST	B
CML254 ^a	ST	E	NC350	ST	B
CML258 ^a	ST	B	NC352	ST	B
CML261	ST	C	NC354	ST	B
CML277	ST		ND246	NSS	C, E, F, H
CML281	ST	B	Oh7B	NSS	
CML287	ST	B, C, E	Oh43 ^a	NSS	C, E, G
CML333 ^a	ST	C, F, H	P39 ^a	NSS	D
CMV3	NSS	B	Pa91 ^a	NSS	
D940Y ^a	NSS	B	Q6199	ST	B, E
EP1 ^a	NSS	B, E, F, H	SA24	NSS	B, D
F2 ^a	NSS	E	SC55	ST	B, G
F7	NSS	C, E	SC213	Mixed	B, C
F44	NSS	E	Sg18	NSS	
F2834T	NSS	C, F, H	T8	ST	F, G, H
GT112	NSS	C, F, H	T232 ^a	NSS	
H95	NSS	G	Tx601 ^a	ST	C
H99	NSS	C	Tzi8	NSS	
HP301	NSS	C	Tzi10	ST	B, E, F, H
I-29 ^a	Mixed	B, E, G	Tzi18	ST	B, E
I137TN	NSS	H	U267Y	NSS	B, C
I205 ^a	NSS	C	Va26	NSS	E
Ia2132 ^a	NSS	F, H	W64A	NSS	C
IDS28 ^a	NSS	D	W117Ht	NSS	D, E
IL14H	NSS	C	W153R ^a	NSS	E
IL101 ^a	NSS	C	W182B	NSS	C, E
IL1677a	NSS	C	Wf9	NSS	

The subpopulation classification is based on Remington et al. (2001). Subpopulations are denoted by the following: NSS, nonstiff stalk; SS, stiff stalk; ST, subtropical/tropical. Allele designations are as follows: B, line containing polymorphism *Sh1*-1210G; C, line containing polymorphism *Sh1*-775C; D, line containing polymorphism *Sh2*-3674-1; E, line containing polymorphism *Bt2*-925T; F, line containing polymorphism *Ae1*-1509G; G, line containing polymorphism *Sh2*-3842G; H, line containing polymorphism *Ae1*-1689C.

^a Member of the 32-line subset for which all six genes (except *ae1*) were sequenced over their entirety.

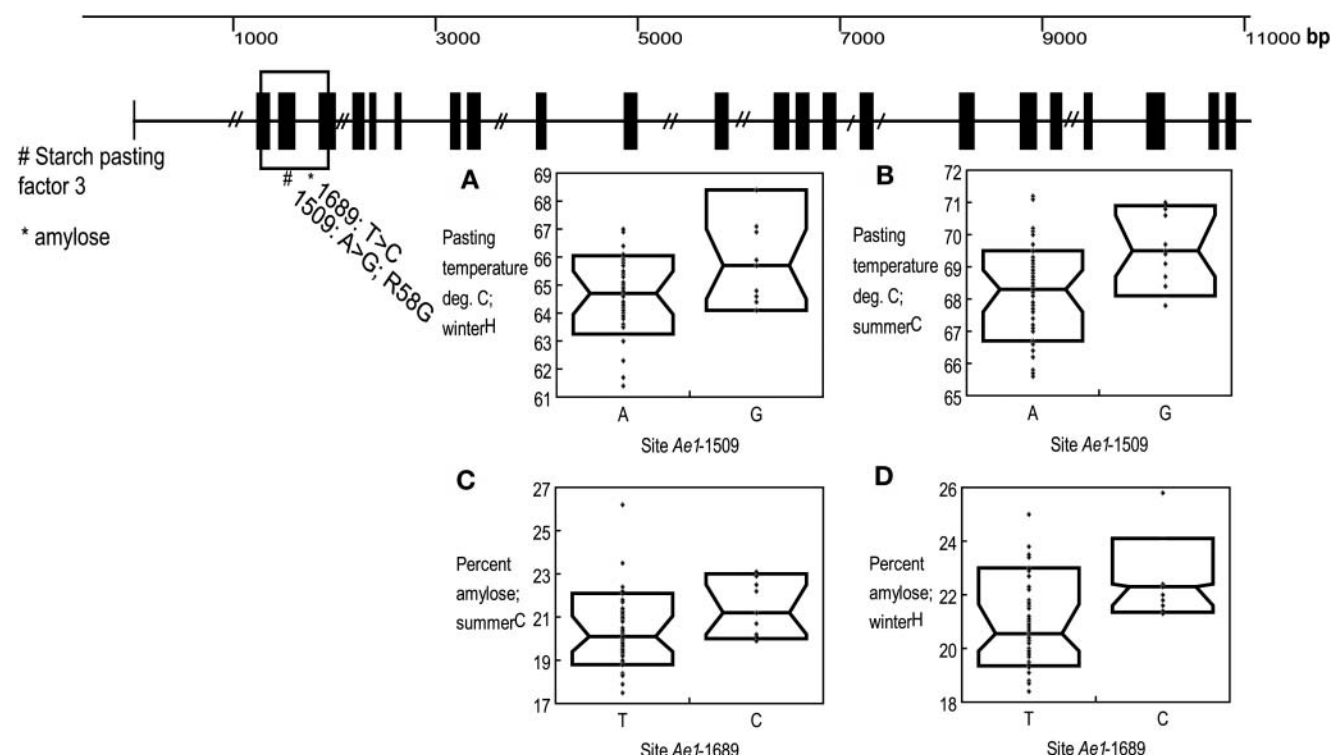


Figure 2. Genetic Structure of *ae1* Sequenced from 32 Maize Taxa.

The gene area outlined with a box denotes the region sampled in the full set of 102 taxa. Sites significant for starch pasting factor three (#) and amylose (*) are highlighted. Allele *Ae1*-1509G associated with higher pasting temperatures in both the summer^C and winter^H replications (**[A]** and **[B]**). Allele *Ae1*-1689 associated with higher amylose in both summer^C and winter^H replications. Insets show the distribution of the data: the median for the data points is marked by the middle horizontal line; the upper and lower horizontal lines highlight the 10th and 90th percentiles. Figures for genes *bt2*, *sh1*, and *sh2* are set up in the same fashion. Note that the diagonal lines (//) in the gene picture are intronic areas and exon 15 not sequenced in the initial 32 lines.

significantly different, the variance in oil content in lines with the *Bt2*-925(T) allele was significantly lower than lines with the more common allele (*F* test; $P < 0.002$) (Figure 3). Nineteen lines out of 102 contained polymorphism *Bt2*-925(T) and were of nonstiff stalk or semitropical origin (Table 3).

sh1

Overall, *sh1* associated with a general $G \times E$ effect (kernel composition factor three; Table 2) (logistic regression; $P = 0.02$). PCA weighted all kernel traits similarly for factor three but in opposite directionality across replications, suggesting the $G \times E$ effect. The most significant site was SNP *Sh1*-1210(A $\leftrightarrow$ G) located in intron one (Figure 4). The *Sh1*-1210(G) allele occurred in 35 out of the 102 lines surveyed, the majority being of tropical origin (Table 3).

sh1 also associated with amylose content in the summer^C replication (logistic regression; $P = 0.002$) (Table 2). The most significant polymorphism was SNP *Sh1*-775(T $\leftrightarrow$ C), which is located within the promoter region (Figure 4). Lines with the *Sh1*-775(C) allele contain 4.6% less amylose relative to the more common allele, *Sh1*-775(T) (GLM; $P = 0.0001$). Forty-one lines

out of 102 contained the *Sh1*-775(C) allele, and it explained $\sim 12\%$ of the total variation. All three subpopulations have representative lines containing allele *Sh1*-775(C) (Table 3); however, the stiff stalks had the highest frequency of *Sh1*-775(C) at 75%. The amount of LD between *Sh1*-775(T $\leftrightarrow$ C) and other polymorphic sites in *sh1* was low (LD; $r^2 < 0.30$).

sh2

Overall gene analysis for *sh2* indicated a significant association with a general $G \times E$ effect (kernel composition factor three; Table 2) (logistic regression; $P = 0.035$). The most significant polymorphism included a 1-bp deletion in intron eight at site 3674, *Sh2*-3674(INDEL1) (Figure 5). Examination of the sequence alignment for the entire gene from all 32 lines revealed that allele *Sh2*-3674(del1) is in LD with a suite of other polymorphisms, including multiple SNPs, an 8-bp insertion in the promoter (581 bp away from the noncoding exon one), an 11-bp deletion located in intron 10, and a 67-bp deletion in intron 13 (site 4640), indicating an obvious haplotype. Allele *Sh2*-3674(del1) occurred in 10 out of the 102 lines surveyed, mainly

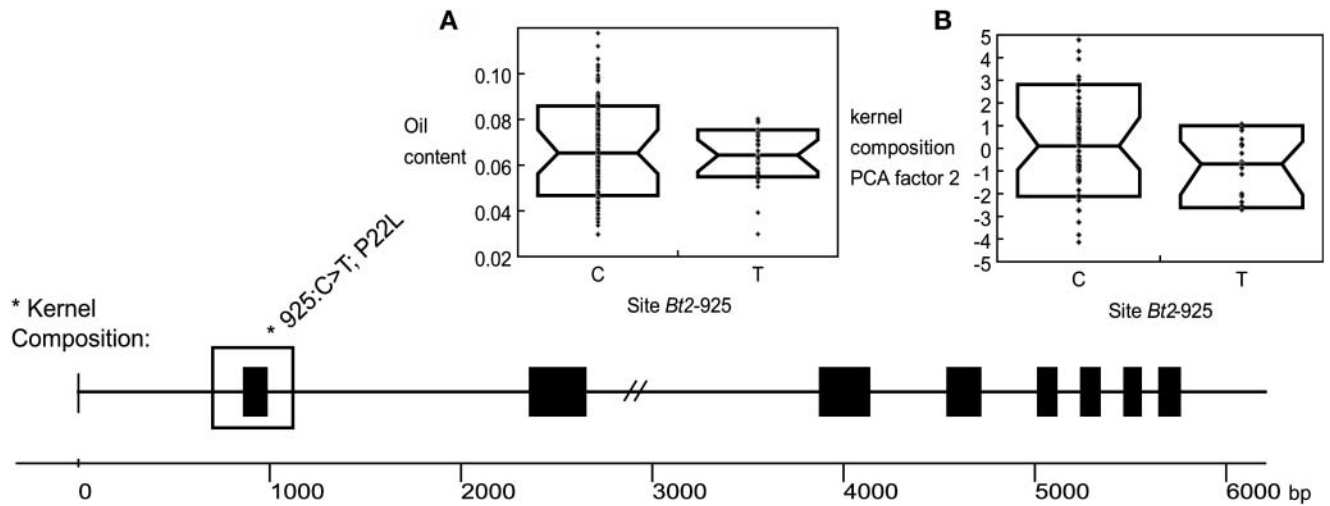


Figure 3. Genetic Structure of *bt2* with the Significant Region Associated with Kernel Composition Factor Two for Oil.

The last exon of *bt2* (exon nine) was not sequenced. Diagonal lines near position 3000 (//) denote an area not sequenced because of a highly repetitive 250-bp stretch.

in nonstiff stalks (Table 3). This same polymorphic site in *sh2* also showed an overall association with starch viscosity characteristics (summer^C starch pasting factor one; Table 2) (logistic regression; $P = 0.05$).

Because *sh2* contained several clearly defined polymorphic alleles, haplotype analysis was performed to increase power to detect associations. Six haplotypes based on six polymorphic sites were tested for associations with seven viscosity traits

(breakdown, consistency, cool paste viscosity, hot paste viscosity, peak viscosity, setback, and trough viscosity) and with amylose content, using GLM with population structure (Figure 6). *sh2* haplotypes showed significance mainly in the summer^C replication, where associations were seen with consistency, cool paste viscosity, hot paste viscosity, peak viscosity, and trough viscosity (Table 4). Significant associations were seen with amylose content and starch breakdown in the winter^H

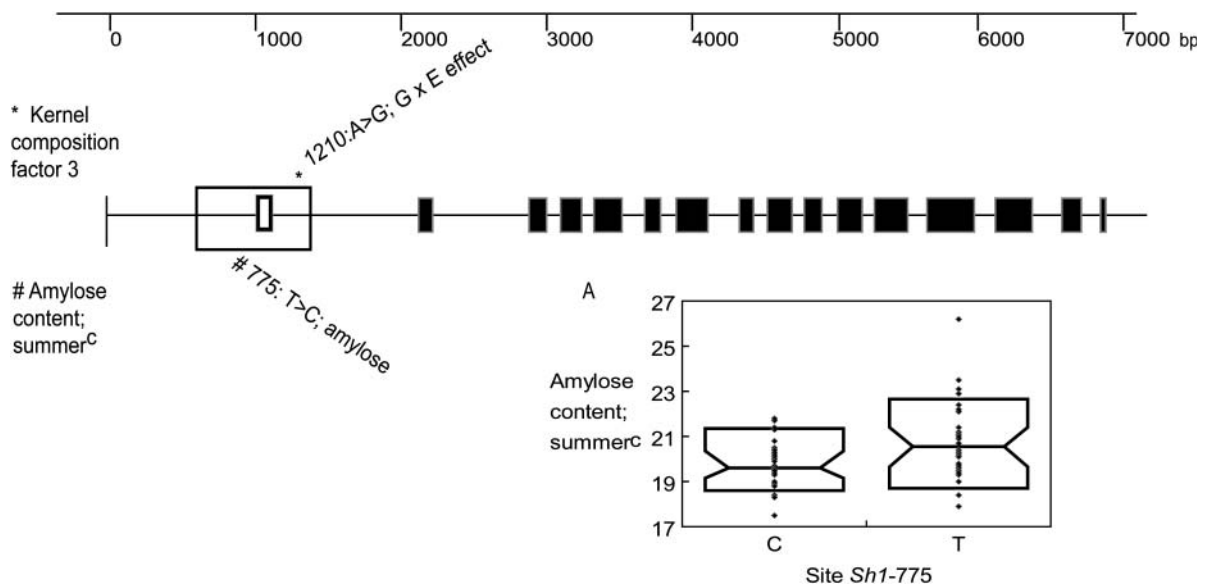


Figure 4. Genetic Structure of *sh1* and the Region Significantly Associated with Kernel Composition Factor Three for $G \times E$ Effects and for Amylose Content.

The black, unfilled exon denotes noncoding exon one of *sh1*.

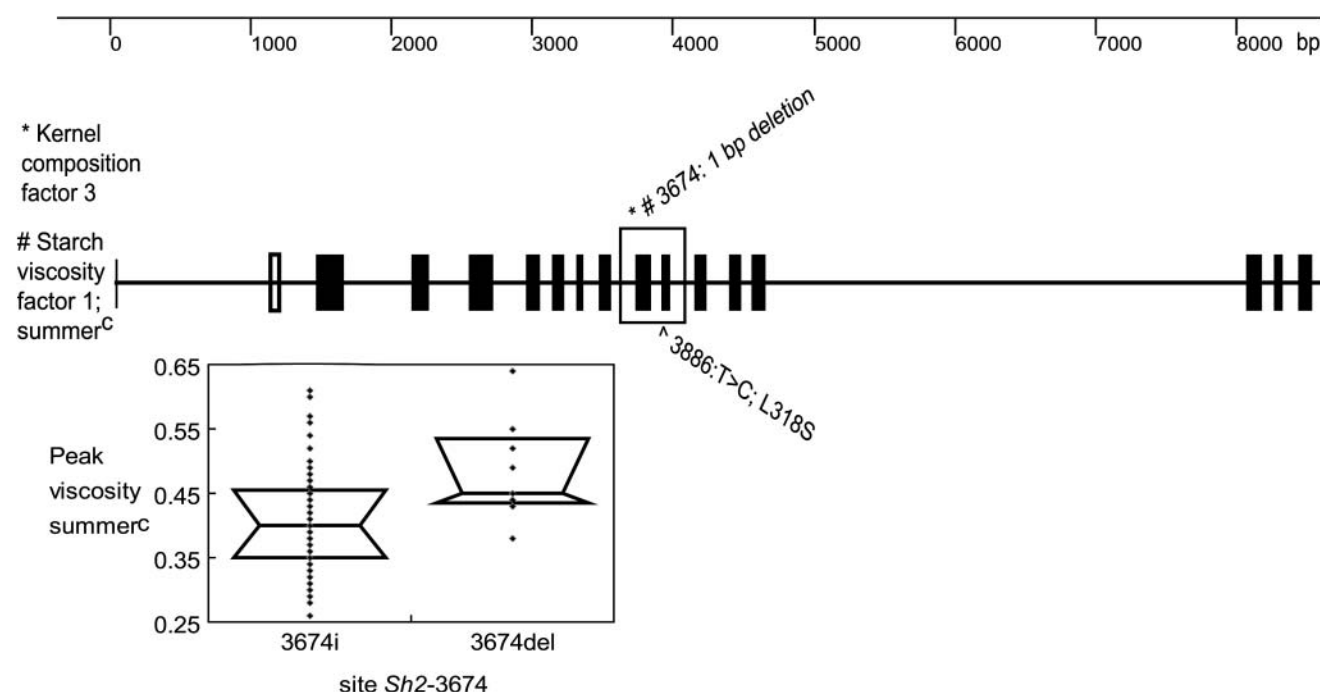


Figure 5. Genetic Structure of *sh2* and the Region Significantly Associated with Kernel Composition Factor Three for $G \times E$ Effects and with Starch Pasting Factor One for Multiple Viscosity Traits.

Only the trait peak viscosity is shown for space conservation for the allelic distribution of the data.

environment. Furthermore, a comparison of *sh2* haplotypes was performed between winter^H and summer^C replications for mean amylose content and kernel composition factors one through three (Figure 6). Haplotype A containing the *Sh2*-3674(del1) polymorphism (which associated significantly with both kernel composition and starch pasting traits) had the largest negative weighting with kernel composition factor three (the $G \times E$ factor) and one of the largest negative effects on amylose content. A different *sh2* haplotype, B, had the greatest (positive) least squares mean in amylose content compared with the other five haplotypes. This B haplotype contains the most significant polymorphism in the overall gene analysis, polymorphism *Sh2*-3842(G), which was marginally significant for amylose content in winter^H (logistic regression; $P = 0.095$) (Table 2). Significance increased using haplotype analysis, where the strength of an association with amylose content was $P = 0.01$ (Table 4).

The Sene et al. (2000) QTL study found a significant effect on amylose content that colocalized with *sh2*. Their study compared genotypes F2 and I205 (I205-like). Genotype I205 contains the most common haplotype in the sample, C, whereas genotype F2 contains haplotype D. Although these two haplotypes were significantly different for kernel composition PC3 ($P = 0.03$), they were not significant for amylose content in summer^C. One particular polymorphism of interest found within genotype F2, but not in the I205 haplotype, is a Leu to Ser amino acid substitution at residue 318. Polymorphism L318S is also found in a second haplotype, E, where member lines contained a high percentage of amylose. When haplotypes D and E containing

L318S were compared against the I205 haplotype C, a significant difference in kernel composition PC3 was seen.

DISCUSSION

Six major genes controlling starch content in maize were analyzed using an association approach to identify those haplotypes affecting starch content or quality, which could be used in maize improvement. Mutational studies have already shown that *ae1*, *bt2*, *sh1*, *sh2*, *su1*, and *wx1* have major effects on either the amount of starch produced or the type, as indicated by different amylose or amylopectin levels.

Phenotypic traits were organized into three major groupings for this analysis: kernel composition, starch pasting properties, and amylose content. Significant kernel composition associations were found in the three starch production genes *sh1*, *sh2*, and *bt2*. Tests for association with either starch pasting characteristics and/or amylose content were significant in *sh1*, *sh2*, and *ae1*.

Resolution and Power Implications for Detecting Associations

Because association approaches rely on the variation produced by evolution for mapping, the evolutionary history of a particular locus affects both genetic resolution and statistical power. The genetic resolution of association approaches is directly dependent upon the structure of LD, and generally the rapid decay in

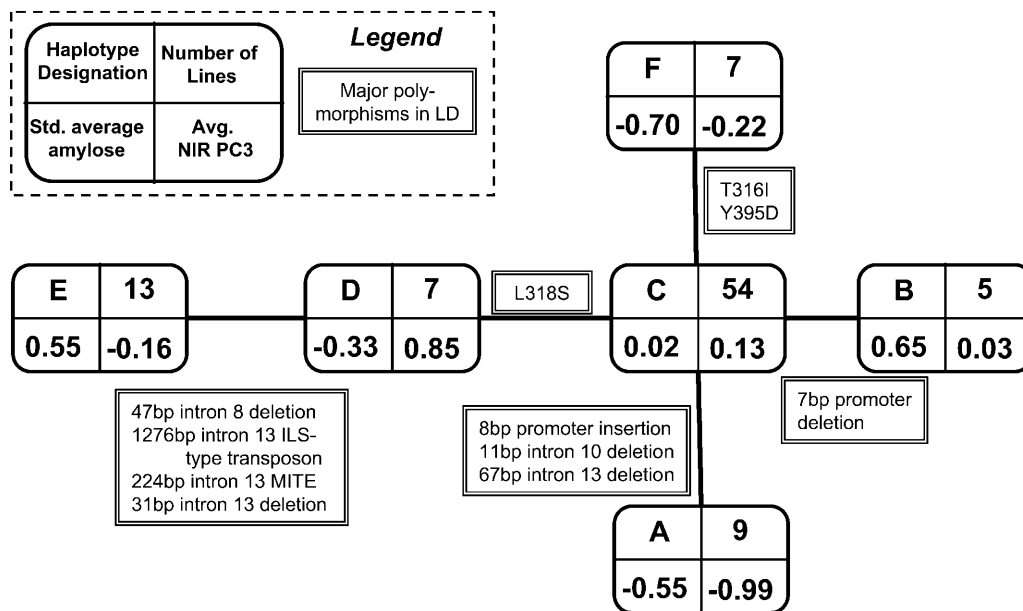


Figure 6. *sh2* Haplotype Network.

Sequencing of a 500-bp region of *sh2* revealed six major haplotypes, with haplotype C being the most common with 54 member lines. The least squares mean for amylose content (both winter^H and summer^C replications) and kernel composition factor three (the G × E factor) are reported for each haplotype, signifying the haplotype's average effect on these traits. Major polymorphisms of interest and in LD occurring outside the analyzed region of *sh2* are indicated along the branches where the mutations likely occurred.

association populations provides high resolution. Prior work using the same diverse maize panel at *Dwarf8* revealed rapid LD decay, and associations were not seen in genes within 1 cM (Thomsberry et al., 2001). By contrast, LD at the *Y1* locus—with its near perfect penetrance and the target of very strong selection in this last century—can be more extensive (Palaisa et al., 2003). Among the six genes studied here, three exhibit LD decay within 2000 bp, whereas the remainder had slower LD decay (although still within a gene's length). This lack of decay is almost certainly because of diversity reductions caused by selection (Whitt et al., 2002). Ancestral recombination can still be retained after some selection bottlenecks, however, as evidenced by the rapid LD decay of *bt2* despite its being a target of selection. Although LD decay varied among the six genes studied here, the extent of LD decay and, thus, the resolution of this association approach still exceeded that of linkage mapping by several orders of magnitude.

In terms of statistical power, association approaches can be limited by several factors. (1) Allele class frequency directly affects the statistical power of all mapping approaches. In a standard linkage population with two inbred founders, alleles have a frequency of 50% (excluding segregation distortion), whereas in an association study, allele frequency can be highly variable. As such, there will always be less statistical power to evaluate rare alleles in an association study. Our findings revealed few associations for those genes with low diversity (*ae1*, *bt2*, and *su1*), which may result from a lack of power to test rare alleles. However, because selection events can skew allele fre-

quencies (with deleterious alleles becoming very rare or extinct), perhaps all functional variation was eliminated from these loci as a result of selection—leaving no associations to detect.

(2) The number of alleles per locus must also be considered when detecting associations because more alleles require more statistical tests per locus. In this study, we performed gene-wise P-value corrections, resulting in a higher stringency per SNP for more diverse loci. Most of the robust associations, however, occurred in *sh1* and *sh2* (logistic regression; $P = 0.002$ and 0.02 overall, respectively)—two of the three most diverse genes in this survey. Although greater numbers of polymorphic sites will decrease statistical power, it is encouraging to note the strength of the associations identified here for several diverse genes for traits with modest heritability.

(3) The interactions between diverse alleles across the genome may have limited the statistical power needed to evaluate some functional polymorphisms. Although the extremely diverse nature of the germplasm used in this study no doubt ensured that diverse alleles with large effects were present in the population, differences in days to flower make it likely that only those alleles with consistent effects across the season were detected. Although such diversity in grain-filling times does not cause false positives (as evidenced by our empirical test using genes in unrelated pathways), it may reduce the power needed to detect associations.

To address these issues, statistical power can be improved by increasing sample size. This not only increases the possibility of capturing more instances of less common alleles but also

Table 4. Results of *sh2* Haplotype Analysis

Trait	Field Replication P-Values	
	Winter ^H	Summer ^C
Amylose	0.010	0.287
Breakdown ^a	0.006	0.367
Gel consistency ^b	0.524	0.054
Cool paste viscosity ^c	0.573	0.014
Hot paste viscosity ^d	0.480	0.008
Peak viscosity ^e	0.174	0.041
Setback ^f	0.556	0.065
Trough viscosity ^g	0.541	0.004
Pasting PC1	0.415	0.025
Pasting PC2	0.185	0.680
Pasting PC3	0.961	0.801

Highly weighted pasting traits in PC factor one were tested individually for associations. Significant test results are in boldface.

^aBreakdown: peak viscosity minus trough viscosity, indication of breakdown in viscosity of paste during 95°C holding period (Pa-s).

^bGel consistency: cool paste viscosity minus hot paste viscosity (Pa-s).

^cCool paste viscosity: viscosity of paste cooled to 50°C (Pa-s).

^dHot paste viscosity: final viscosity after cooking at 95°C (Pa-s).

^ePeak viscosity: maximum viscosity recorded during heating and holding cycles (Pa-s).

^fSetback: cool paste viscosity minus trough viscosity, indication of retrogradation of cooked rice during cooling (Pa-s).

^gTrough viscosity: minimum viscosity after peak (Pa-s).

provides avenues for dealing with the confounding effects of epistasis. In this study, if the number of genotypes was increased from 100 to 300 or 400, not only could many rare or weaker alleles be evaluated (Long and Langley, 1999) but lower Type I error rates could also be set, resulting in lower false discovery rates. Increasing the sample size would also permit the inclusion of interaction terms in the models, separate statistical tests for each population, or use of specific polymorphisms with known effects as cofactors. Although increasing sample size is statistically straightforward, empirical researchers must overcome the limitations posed by phenotyping many genotypes in replicated environments.

In addition to increased sample size, testcross evaluations (cross all the diverse lines to a tester line and then evaluate the F1 offspring) may also provide an efficient means of reducing genetic background interactions among the lines. Even with the dilution of additive effects that is expected for such testcrosses, the reduction in epistasis that is also achieved may improve the evaluation of additive effects.

Evaluating QTL with Linkage and Association Analysis

QTL mapping studies for starch traits using recombinant inbred line (RIL) or F2 populations have traditionally identified regions that span 10 to 20 cM, corresponding to as many as 20 million bases. By contrast, this study used association analyses to identify a suite of polymorphisms within a few thousand bases, resulting in a substantial increase in resolution. This high reso-

lution raises the question of whether candidate gene association analysis should replace mapping with RIL or F2 populations. A comparison of this association study with RIL mapping suggests that a combination of both these methods provides the most powerful experimental approach for the following three reasons.

First, there is an important tradeoff between statistical power and resolution for all mapping approaches. In standard linkage populations, the equivalent of 50 independent regions are surveyed, whereas in a high-resolution association panel (as in this study) >50,000 such independent regions must be surveyed. Although this may become feasible on a molecular level in the near future, it would still require the phenotyping of a vast number of lines to scan the genome. Genome scan power estimates for alleles with modest effects may require thousands to tens of thousands of individuals (McGinnis et al., 2002). Phenotyping of distinct genotypes would be limiting.

Second, candidate genes with an established position under QTL peaks exhibited more associations than did those outside such peaks. For example, 7 of 11 associations were found in those two genes (*sh1* and *sh2*) that colocalized with QTL for kernel traits, a result that is significantly more than expected (χ^2 test; $P = 0.03$). This suggests that RIL mapping is an efficient way to eliminate the first 90% of the genome.

Third, although the use of RIL and F2 populations in QTL studies tests a maximum of only two alleles, RIL and F2 populations do have substantial statistical power to contrast these alleles. To illustrate this point, associations found in *sh2* in this study were compared with a QTL study done by Séne et al. (2000) that found an effect on amylose content near *sh2*. Although our study was able to confirm that these two alleles have an environment-dependent effect on kernel quality, we were unable to confirm the specific amylose effects, partially because of genotype by environment interactions. We identified a candidate polymorphism, L318S, as a possible basis for the results of both this and the Séne study. Larger populations and, therefore, more statistical power are probably needed to effectively apply association approaches for those genes with numerous alleles, as seen in *wx1*.

Overall, RIL populations can powerfully contrast pairs of alleles with low resolution, whereas association analysis provides a high-resolution evaluation of numerous alleles with uneven statistical power. Exploiting the complementary strengths and weaknesses of both approaches should allow efficient QTL evaluation of the genome.

Potential Functional Polymorphisms

Our association analysis was successful in locating regions and haplotypes that affect maize kernel composition. This has suggested several specific biochemical hypotheses that should be tested in the future: (1) a *sh1* polymorphism located in intron one associated with a $G \times E$ effect—this intron has been related to gene activity (Vasil et al., 1989; Clancy et al., 1994; Clancy and Hannah, 2002). (2) Allele *Bt2-925(T)*, located in *bt2*'s exon one and causing a P22L polymorphism, associated with a decrease in variance in oil content. Genotypes with the *Bt2-925(T)* allele resulted in reduced variability by cutting out the high and low extremes in oil production. (3) *sh2* is the other AGPase gene with

Table 5. Sequence Context of the Major Polymorphisms Identified through the Association Analysis

Polymorphism Name	Sequence (5'/3')
<i>Ae1</i> -1509A>G	GCGCGGCGGCCGCGGCC[A/G]GGAAGCGGTCATGGTTCCT
<i>Ae1</i> -1689T>C	CCTCTTT[T/C]TGGATGCTATTTGAGAACAA
<i>Bt2</i> -925T>C	CGCCGAGCAGC[T/C]AATCCAAAGCGTGACAAAGCCGCTG
<i>Sh1</i> -775T>C ^a	GGTCTGAAC[T/C]TT[T/C]CCGAAACCAGCCAGCCATTGGTCT
<i>Sh1</i> -1210A>G ^b	ATCTGCTGG[T/C]C[C/G]CGGTAGAAAAGA[T/C]C[A/G]TGTCCC
<i>Sh2</i> -3674-1 ^c	{[T/-]ACTGAT}GTTGCAGAGAGTTGAGACCAACTTCCTGAGCT
<i>Sh2</i> -3842T>G	TCTATCCAACCTCCTAGT[G/T]TACCTTCTAACAGTGT

^a The first polymorphic site, [T/C], is the significantly associated site.

^b The significantly associated site is the last polymorphic site, [A/G].

^c The significant polymorphism is the 1-bp deletion indicated by [T/-]. A third allele included lines that had a 47-bp deletion that encompassed this area. The sequence in braces is the last part of this deletion and therefore is missing in these lines.

significant associations—out of the six *sh2* haplotypes examined (Figure 6), two in particular showed interesting effects on pasting traits. Haplotype A, which contains deletion polymorphism *Sh2*-3674(del1), not only seems to have different effects on kernel composition traits under different environmental conditions, but also may have a negative effect on amylose. Furthermore, the A haplotype also displays the highest means over the other five haplotypes for significant viscosity traits in summer^C: consistency, cool paste viscosity, hot paste viscosity, peak viscosity, and setback (data not shown). (4) Variations in *ae1*, a branching enzyme gene, are likely to have an effect on amylose content and/or pasting properties. One particular polymorphism in exon two, *Ae1*-1509(G), caused the nonsynonymous change R58G in the predicted protein sequence. The phenotypic effect seen with this mutation was higher pasting temperatures. The R58G mutation occurred within the identified transit peptide and is located near the cleavage site in front of the N terminus of the mature BEII protein (Fisher et al., 1993). Although an increase in amylose content was seen in lines with the *Ae1*-1689(C) polymorphism, in LD with site 1509, it is unclear what causes an increase in pasting temperature, an important indicator of stability in food processing of starches. Amylose content and pasting temperature did not significantly correlate (data not shown); however, these traits could be related to changes through physicochemical properties of the amylopectin branch lengths, thereby contributing to the stability of the starch under stress.

It should be noted that although the above polymorphisms were the most suggestive functionally, there were in almost all cases other polymorphisms with equal statistical support. Additionally, although large portions of the genes and promoter regions were sequenced for the test associations, we cannot rule out closely linked regions based on the structure of LD at a given locus.

Genotype by Environment Interactions

The multiple occurrences of starch genes that associate with genotype by environment interactions in kernel composition are hardly surprising for a moderately heritable pathway, diverse germplasm, and diverse environments. Of the four field locations where kernels were phenotyped for kernel composition traits,

West Lafayette, IN (2000) displayed poorer grain quality because of a stressed environment in comparison with the other environments. G × E associations with kernel composition were seen in both *sh1* and *sh2*.

sh1 associated significantly with a G × E effect at an intronic polymorphism, *Sh1*-1210 (see above), whereas *sh2* associated significantly with a G × E effect at a 1-bp deletion at position 3674; however, this particular polymorphism was in significant LD with a suite of polymorphisms located throughout the entire gene to form a limited number of haplotypes, thus limiting resolution. Therefore, the causative polymorphism may not even be located within the *sh2* gene, but reside instead in closely associated flanking regions. This latter association may involve the SH2 subunit of the AGPase enzyme. Evidence for AGPase suggests that alternate alleles produce enzymes that perform differently in a stressed environment, affecting factors such as heat lability or altering SH2:BT2 interactions. Mutations in the SH2 subunit have been shown to increase its stability, thereby increasing SH2:BT2 interactions (Greene and Hannah, 1998).

The G × E nature of *sh2* may also be reflected in the results of viscosity associations (Table 4). Haplotype analysis of *sh2* allowed for a more powerful examination of the pasting traits driving PC1. In the winter^H replication, haplotypes differed significantly in amylose content and starch breakdown, but not in the summer^C environment. Conversely, in the summer^C environment, almost all remaining viscosity traits thought to drive PC1 (consistency, cool paste viscosity, hot paste viscosity, peak viscosity, and trough viscosity) were significant by *sh2* haplotype but were not significant in the winter^H replication. Therefore, *sh2* haplotypes may be showing G × E effects in viscosity traits as well, but more than two replications are needed to obtain adequate statistical power to detect this. Differing associations between summer^C and winter^H environments on downstream traits, such as starch pasting, may be influenced by the results we obtained for the G × E nature of *sh2* on general kernel composition traits.

One commonality this study shared with the Séne study (Séne et al., 2000) is that the effect of *sh2* on amylose was not seen in all environments. Because *sh2* showed a significant G × E effect on kernel composition, we propose the hypothesis that with higher ambient temperatures, the heat labile nature of AGPase (Greene

and Hannah, 1998), in which *sh2* encodes a subunit, varies between *sh2* haplotypes and has an epistatic effect on amylose production. When the two haplotypes of F2 and the Iodent I205 were compared, significance with the $G \times E$ effect on kernel composition was also seen, further explaining the inconsistent results for amylose between environments.

Population Structure and Association Analysis

Although *sh2* has a significant effect on the amylose/amylopectin ratio in certain environments, a previously reported association with overall starch may be a false positive result. Prioul et al. (1999) found an association with a *SacI* restriction site within the *sh2* gene in a sample of 46 unrelated maize inbreds but did not control for population structure. Our analysis also finds a significant association if population structure is ignored because tropical, stiff stalk, and nonstiff stalk germplasm all have different mean starch levels. Although the Prioul association may reflect the $G \times E$ nature of *sh2*, another possibility is that this association with starch is purely a result of population structure. Population structure was significantly related to basic kernel composition (PC1) (GLM; $P < 0.02$) and accounted for 10% of the variation. Because kernel quality is correlated with population structure, associations performed without a population structure correction need to be reevaluated. The particular story at *sh2* may also be complicated by a tightly linked locus to *sh2* with effects on starch, as suggested by a recent QTL study based on crosses of the Illinois high and low protein maize strains that not only found a QTL at *sh2*, but also more strongly at a linked marker (Dudley et al., 2004).

To effectively use association approaches in the study of plant genetics, population structure must be considered to prevent false positive results (Knowler et al., 1988; Pritchard et al., 2000a, 2000b). By controlling for population structure, Thornsberry et al. (2001) were able to locate polymorphisms within the *Dwarf8* gene that associated with flowering time variation in maize. In this study, estimates of population structure for the diverse set of inbred lines, as determined in Remington et al. (2001), were incorporated into all analyses. This allowed for the detection of significant associations by increasing power through the use of an unlinked set of markers. In some instances, however, correcting for population structure actually caused several genes to lose significance (data not shown). This loss of significance can have two causes: (1) this was a nonfunctional polymorphism and the association was caused by population structure, or (2) the polymorphism is functionally related but the polymorphism distribution coincides with population structure. The second case results in a functionally false negative result. If this is the case, then the polymorphism needs to be reevaluated in alternative population structures.

Implications

To date, only a handful of QTL have been dissected to the gene level in plants. By building on previous linkage mapping populations, this study used association approaches to identify at least three additional genes with QTL effects. This study has supplied breeders with a set of high-resolution markers for a set of six starch genes. Ultimately, these markers can be used to meet

specific starch or yield goals by incorporating desirable alleles into maize germplasm. This study has also provided a wealth of candidate polymorphisms for future analysis by molecular biologists and biochemists to further elucidate this critical plant pathway. As previously acknowledged, the modest sample size perhaps limited power to detect all associations in such a diverse germplasm. By reevaluating these candidate polymorphisms in larger association populations and in testcrosses, additional functional polymorphisms may yet be discovered. As such, this study represents a promising beginning to evaluating functional nucleotide diversity in the maize starch pathway.

METHODS

Plant Materials

The 102 maize inbreds (*Zea mays* ssp *mays*) used in this study represent most of the diversity available to breeding programs around the world, retaining at least 80% of the microsatellite diversity found in maize landraces (Liu et al., 2003). These inbreds were divided into three subpopulations based on simple sequence repeat data described by Remington et al. (2001): the stiff stalks, the nonstiff stalks, and the subtropical/tropicals (Table 3). Note that some lines traditionally known as stiff stalk lines (B37, for example) are grouped with the nonstiff stalk lines, and the one white popcorn in the study (I29) is grouped with the tropicals here. The inbreds were grown in one-row plots in a randomized complete block design at four different field sites in the U.S., with a total of six replications: Homestead, FL (winter 1998 to 1999); West Lafayette, IN (summer 2000); Clayton, NC (summer 2001) in two replications; Urbana, IL (summer 2001) in two replications. Ten to fifteen plants were self-pollinated by hand in each row; ears were then harvested at maturity and dried and shelled.

Starch Isolation and Phenotyping

Approximately 10 grams of seed from each inbred was pooled from several ears and ground using an M-2 Stein Mill for 90 s. Kernel starch, oil, protein, and moisture percentage were measured from ~600 mg of ground sample using a Dickey-john GAS III NIR light reflectance machine (Hymowitz et al., 1974). All six replications were phenotyped by NIR.

Maize starches were isolated by salt steeping (Rani and Bhattacharya, 1995) and freeze dried. Starch true amylose content was determined in triplicate with an amylose/amylopectin assay kit (K-AMYL; Megazyme International, Wicklow, Ireland) following a simplified concanavalin A procedure (Gibson et al., 1997).

Starch pasting properties of maize starches were determined using a controlled stress rheometer (AR 1000-N, Rheolyst; TA Instruments, Dover, DE) at a constant shear rate of 200 s^{-1} . The rheometer was fitted with a polysulfone cone, which had a diameter of 4 cm and an angle of 4° . A microviscoamylographic method, which only requires 100 mg of sample, was used. The temperature program used consisted of four segments: (1) 45 to 95°C in 3.5 min, (2) holding at 95°C for 2.3 min, (3) 95 to 50°C in 3.5 min, and (4) holding at 50°C for 1.25 min. Triplicate aqueous maize starch suspensions (8% w/w) were prepared using deionized water. Suspensions were degassed while stirring under vacuum for 15 min (Ibáñez, 2002). The definitions for measured properties include the following: pasting temperature, temperature of initial viscosity increase; peak time, time required to reach peak; peak temperature, temperature of peak viscosity; peak viscosity, maximum viscosity recorded during heating and holding cycles; trough, minimum viscosity after peak; hot paste viscosity, final viscosity after cooking at 95°C ; cool paste viscosity, viscosity of paste cooled to 50°C ; breakdown, difference (–) between peak viscosity and trough, indicating breakdown in viscosity of paste

during 95°C holding period; setback, difference (–) between cool paste viscosity and trough; gel consistency, difference (–) between cool paste viscosity and hot paste. The controlled stress rheometer was operated using TA navigator software (version 4.0; TA Instruments) for a personal computer. For true amylose content and starch pasting properties, kernels from the environments Homestead, FL (1998) and Clayton, NC (2001; one replication) were phenotyped. Sixty-seven nonsweet lines were scored from Homestead, and 90 lines were scored from Clayton; discrepancy in number of samples phenotyped by NIR analysis and starch pasting assays were a result of limited samples available for the starch pasting assays after NIR measurements.

Amplification and Sequencing

The six candidate genes and promoter regions were amplified and direct sequenced in 32 lines as described previously by Whitt et al. (2002). Candidate genomic regions either from the coding region and/or promoter regions were chosen based on the position of amino acid changes and substantial indel polymorphisms. Selected regions were sampled in a set of 102 maize inbreds as follows: for *ae1*, exon one through exon three, exon 12 through exon 14, and exon 16 through exon 18; for *bt2*, the promoter through intron one; for *sh1*, the promoter through the noncoding exon one and a portion of intron one; for *sh2*, intron eight through intron 10; for *su1*, the promoter through exon one and exon 13 through intron 14; for *wx1*, exon one through exon two and exon eight through exon nine. SNP/indel positions referred to in the text correspond to alignment positions from sequences submitted by Whitt et al. (2002). Major alleles are defined in Table 5.

Statistics

To summarize the data over multiple replications, PCA of the correlation matrix was performed on both the NIR and the starch pasting data using SAS software (SAS, 1999) for 97 nonsweet taxa from 102 maize inbreds. Although sequence alignments for the six candidate loci and phenotypic data from five sampled sweet maize lines were made and are available (la2132, ll14H, ll101, ll677a, and P39), these low-starch mutants were removed from the association analyses to avoid false results produced by these outliers (the sample number of sweet lines was too low to form its own subpopulation). Three separate PCAs were performed: one for total NIR data and one each for starch pasting data from the winter (Homestead, FL 1998) and from the summer (Clayton, NC 2001) replications, referred to hereafter as winter^H and summer^C, respectively. To handle missing NIR data for the PCA, missing data were imputed using the KNN impute program (Troyanskaya et al., 2001) with K set to five nearest neighbors. The major principal components (PC) found for NIR and for each starch pasting analysis were then used as traits in association tests for the 97 taxa. Amylose content for both summer^C and winter^H replications, not included in the PCAs, was used directly to test for associations. Starch pasting and amylose data were not pooled from the two environments, as this would have required excessive imputation.

Tests for association (Thornsberry et al., 2001) and LD (Hill and Robertson, 1968) were performed using the software package TASSEL, available at <http://www.maizegenetics.net>. SNPs or indels at a site frequency of 0.05 or greater among the 97 inbreds were evaluated using TASSEL. All association tests were run with population structure included, using logistic regression as described by Thornsberry et al. (2001). One thousand permutations of the data were run to account for multiple tests within a gene, and a significant association was called if the P-value of the most significant polymorphism in a region was seen in <5% of the permutations.

To examine possible allelic effects of significant polymorphisms, post hoc statistical tests (GLM) were used to further dissect PC-associated

effects in those genes with significantly associated PC traits. These tests were used to determine whether the sample means of actual trait values were significantly different between lines with the best site polymorphism in an association and those lines without the polymorphism. GLM models in SAS (SAS, 1999) included estimates of population structure, and reported P-values are from the Type III sum of squares (i.e., the effect after the population structure has been removed).

Sequence data for the six genes for the 102 genotypes included in this article have been deposited with the EMBL/GenBank data libraries under the following accession numbers: *ae1* (AY290043 to AY290190 and AY290192 to AY290304), *bt2* (AY290600 to AY290700), *sh1* (AY290403 to AY290503), *sh2* (AY324882 to AY324981), *su1* (AY290305 to AY290599), and *wx1* (AF544068 to AF544099).

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